

Delegated Entry Threshold

Under agentive delegation, the developer delegates a share k of execution to the AI agent.

The certainty - equivalent surplus from delegated production is:

$$V^D = w + (1-k)S\bar{\mu} + k\alpha z(A) - p(a,s) - r_D - b - \frac{\gamma}{2} \left[\frac{(1-k)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$$

Here, V^D

represents the net value of producing in the language using agentive delegation

Delegated production is feasible when

$$V^D \geq 0$$

Therefore,

$$w \geq b - (1-k)S\bar{\mu} - k\alpha z(A) + p(a,s) + r_D + \frac{\gamma}{2} \left[\frac{(1-k)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$$

Define the delegated entry threshold

$$T^D = b - (1-k)S\bar{\mu} - k\alpha z(A) + p(a,s) + r_D + \frac{\gamma}{2} \left[\frac{(1-k)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$$

Hence,

$$w \geq T^D$$

means that the opportunity value is high enough for delegated production to be profitable

Delegation Advantage and Activation Band

For an unfamiliar language, Assumption 1 implies that conversational AI doesn't lower the entry threshold:

$$T^1 = T^S$$

The solo-production threshold is

$$T^S = b - s\bar{\mu} + \frac{\gamma s^2}{2\pi}$$

Define the delegation advantage

$$B = T^S - T^D$$

B measures how much agentic delegation lowers the entry threshold relative to the pre-agent case.

Substituting T^S and T^D

$$B = b - s\bar{\mu} + \frac{\gamma s^2}{2\pi} - \left[b - (1-k)s\bar{\mu} - k\alpha z(A) + p(a,s) + r_D + \frac{\gamma}{2} \left[\frac{(1-k)^2 s^2}{\pi} + \nabla_D^2(a,s,A) \right] \right]$$

Operating, we obtain

$$B = \underbrace{k[\alpha z(A) - s\bar{\mu}]}_{\text{execution substitution}} - \underbrace{p(a,s) - r_D}_{\text{verification and delegation costs}} + \underbrace{\frac{\gamma}{2} \left[\frac{(2k-k^2)s^2}{\pi} - \nabla_D^2(a,s,A) \right]}_{\text{risk substitution}}$$

If

$$B > 0$$

then

$$T^D < T^S = T^1$$

Therefore

$$T^D \leq w < T^S$$

is the activation band: opportunities that were infeasible before agentic AI but become feasible under delegation

Finally, $Z^2 - Z^1 = 1 \mid T^D \leq w < T^S \mid$

Economic intuition: agentic AI expands the production frontier when delegated execution lowers the entry threshold enough to activate previously infeasible language opportunities.